Digit Convolution Constraints for Integer Factorization: SMT Encodings, Lattice Structure, and Restricted-Model Lower Bounds

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Abstract

We investigate integer factorization through the lens of digit convolution constraints, a formulation that decomposes the problem $n = p \cdot q$ into base-b digit-level arithmetic with explicit carry propagation. This formulation encodes factoring as a system of modular constraints $\alpha_k + t_{k-1} \equiv c_k \pmod{b}$ where $\alpha_k = \sum_{i+j=k} x_i y_j$ is the k-th digit convolution coefficient. We implement this encoding as SMT bitvector assertions in Z3 and compare it systematically against circuit-based SAT encodings and raw bitvector multiplication. A base sweep over 18 bases reveals that power-of-2 bases consistently outperform others, achieving up to 16× speedup over raw bitvector at 32 bits. However, multi-instance experiments (50 balanced semiprimes per bit size) show that the encoding advantage is not statistically significant (Cohen's $d < 0.31$). We then analyze the lattice structure of the carry-propagation system, identifying the rank-1 constraint $Z = \mathbf{x} \otimes \mathbf{y}$ as the precise locus of hardness: LLL efficiently solves the linear subsystem but cannot enforce the nonlinear rank-1 condition. An SDP relaxation of the rank-1 constraint fails with a 30–39% integrality gap. We prove that the carry-propagation lattice contains exactly $2^{d^2/4+O(d)}$ points (base 2), establishing the closed-form constant $\alpha = 1/4$ via a row-sum identity and binomial parity argument, and generalize to $\alpha(b) = \log_2((b-1)^2 + 1)/4$ for arbitrary bases. A spectral analysis reveals a sharp base-dependent landscape: in base 2, the only near-rank-1 lattice points are the two exact factorizations, whereas in base 10, up to 79% of lattice points are spurious near-rank-1, explaining why nuclear norm minimization fails in larger bases. SVD-based blind deconvolution of the carry-constrained system achieves high correlation with true factors (0.83 in base 10) but 0% recovery rate, demonstrating a spectral certification-vs-search gap. Finally, we establish unconditional lower bounds in black-box and generic oracle models, proving that rank-1 points constitute a $2^{-\Omega(d^2)}$ fraction of the carry-propagation lattice, with degree-4 moment indistinguishability providing empirical support for bounded-view rounding hardness.

1 Introduction

1.1 Background

The computational complexity of integer factorization remains one of the central open problems in computer science. While factoring is known to lie in BQP via Shor's algorithm [14], its classical complexity status is unresolved: it is not known to be in P, nor is it known to be NP-complete. The best classical algorithm, the General Number Field Sieve (GNFS), runs in time $L_n[1/3, c] = \exp\big(c(\log n)^{1/3}(\log\log n)^{2/3}\big)$, which is sub-exponential but super-polynomial.

A parallel line of research has explored constraint-satisfaction approaches to factoring, encoding the problem as a Boolean satisfiability (SAT) or satisfiability modulo theories (SMT)

instance. The standard approach constructs a binary multiplication circuit, converts each gate to clauses via Tseytin transformation, and feeds the result to a SAT solver [16]. While conceptually clean, this circuit-based encoding handles only 20–30 bit semiprimes before timing out [12], and remains exponential even with modern CDCL solvers.

Recent hybrid approaches combine SAT solvers with lattice reduction: Ajani et al. [1] showed that when approximately 50–60% of a factor's bits are leaked, a circuit-based SAT solver combined with Coppersmith's lattice method can factor 768-bit numbers in under 800 seconds. Without leaked bits, the approach remains exponential.

1.2 Contributions

This paper makes the following contributions:

1. **Digit convolution formulation.** We present a systematic treatment of integer factorization as digit convolution with carry propagation (Section 2), formulating the problem as modular arithmetic constraints rather than Boolean circuits. The base b serves as a tunable parameter with no analog in circuit-based approaches.

2. **SMT encoding and experimental evaluation.** We implement the digit convolution constraints as Z3 bitvector assertions and conduct systematic experiments (Sections 3–4): a base sweep showing power-of-2 bases dominate; multi-instance benchmarks establishing that the encoding advantage is not statistically significant at 32 bits; scaling law fits confirming digit constraints share the same exponential growth class; and leaked-bit experiments showing a phase transition at 50–70% known bits.

3. **Lattice analysis and the rank-1 barrier.** We reformulate the carry-propagation system as a lattice problem (Section 5) and identify the rank-1 constraint $Z = \mathbf{x} \otimes \mathbf{y}$ as the precise obstruction to efficient solution.

4. **Transfer matrix spectral analysis with closed-form constant.** We prove $\log_2 |\Lambda_n \cap \mathcal{B}| = d^2/4 + O(d)$ for base 2, establishing the exact constant $\alpha = 1/4$ via a row-sum identity and binomial parity argument (Theorem 1). For general base b, we derive $\alpha(b) = \log_2((b-1)^2 + 1)/4$.

5. **Information-theoretic carry channel analysis.** Using the forward–backward algorithm and the Diaconis–Fulman eigenvalue theory [5], we prove the carry chain reveals only $\Theta(d \log d)$ bits of a $\Theta(d^2)$-bit lattice—a vanishing fraction $O(\log d/d)$ (Theorem 4).

6. **SOS rounding hardness.** Degree-4 SOS closes the continuous gap to $< 0.1\%$, but we prove sequential rounding fails with probability $2^{-\Theta(d^2)}$ (Theorem 5). To our knowledge, this is the first explicit certification-vs-search separation demonstrated in a number-theoretic recovery formulation.

7. **Base-dependent spectral landscape.** We prove that base 2 has zero spurious near-rank-1 lattice points ($\sigma_2/\sigma_1 < 0.1$), while base 10 has up to 79% spurious near-rank-1 points, explaining the failure of nuclear norm minimization in larger bases (Theorem 6).

8. **Blind deconvolution experiment.** SVD of the minimum-norm carry-constrained solution achieves average factor correlation 0.83 (base 10) but 0% recovery, demonstrating a spectral certification-vs-search gap.

9. **Moment indistinguishability.** At linear budget ($k = d$ moment entries), 26–43% of semiprime pairs share identical degree-4 moment entries, with agreement increasing with d, supporting bounded-view rounding hardness.

10. **Restricted-model lower bounds.** We establish unconditional lower bounds in black-box and generic oracle models (Section 6), proving that rank-1 points are exponentially sparse ($2^{-\Omega(d^2)}$ fraction) in the carry-propagation lattice.

11. **A Rust implementation** of the backtracking solver achieving 30–33× speedup over Python with identical search paths (Appendix A).

1.3 What This Paper Is Not

We emphasize that this paper does not present a new factoring algorithm competitive with GNFS, ECM, or even Pollard's rho. Classical algorithms outperform our SMT-based approach by over four orders of magnitude at 32 bits. The contribution is a *framework*: digit convolution provides a clean decomposition of factoring hardness into a tractable linear part (carry propagation) and an intractable nonlinear part (the rank-1 constraint), offering a new analytical perspective on why integer factoring is hard.

2 Digit Convolution Formulation

2.1 Base-b Digit Decomposition

Let n be a positive integer to be factored, and let $b \geq 2$ be a base. Write $n = p \cdot q$ where $2 \leq p \leq q$. The base-b digit representations are:

$$p = \sum_{i=0}^{d_x-1} x_i \cdot b^i, \quad q = \sum_{j=0}^{d_y-1} y_j \cdot b^j, \quad n = \sum_{k=0}^{d-1} c_k \cdot b^k \tag{1}$$

where $x_i, y_j \in \{0, 1, \ldots, b-1\}$, c_k are the known digits of n, $d_x = \lceil \log_b p \rceil$, $d_y = \lceil \log_b q \rceil$, and $d = \lceil \log_b n \rceil$.

2.2 Convolution Coefficients

The product $n = p \cdot q$ induces a convolution on the digit vectors. Define the k-th convolution coefficient:

$$\alpha_k = \sum_{\substack{i+j=k \\ 0 \leq i < d_x \\ 0 \leq j < d_y}} x_i \cdot y_j \tag{2}$$

This is exactly the coefficient of b^k in the polynomial product $p(b) \cdot q(b)$ before carry propagation.

2.3 Carry Propagation

Define the partial sum and carry at each digit position:

$$m_k = \alpha_k + t_{k-1} \tag{3}$$

where t_{k-1} is the carry from position $k-1$ (with $t_{-1} = 0$). The digits of n are determined by:

$$c_k \equiv m_k \pmod{b}, \qquad t_k = \lfloor m_k / b \rfloor \tag{4}$$

Equivalently, the system of constraints is:

$$\sum_{i+j=k} x_i \cdot y_j + t_{k-1} - b \cdot t_k = c_k, \quad k = 0, 1, \ldots, d-1 \tag{5}$$

2.4 The Constraint System

Factoring n in base b reduces to finding digit vectors $\mathbf{x} = (x_0, \ldots, x_{d_x-1})$ and $\mathbf{y} = (y_0, \ldots, y_{d_y-1})$ with $x_i, y_j \in \{0, \ldots, b-1\}$ satisfying the carry-propagation constraints (5). The system is *bilinear*: the products $x_i y_j$ appear linearly in the constraints, but the relationship between $\mathbf{x}$, $\mathbf{y}$, and the products is nonlinear.

Historical note. This formulation was first developed by the author in 2009, arising from viewing integer multiplication as polynomial evaluation at b.

2.5 Connection to Polynomial Factoring

If we remove carry propagation (setting all $t_k = 0$), we obtain polynomial factoring over $\mathbb{Z}[x]$. The LLL algorithm [7] solves polynomial factoring in polynomial time because the constraint system is *linear* in the products $z_{ij} = x_i y_j$. Integer factoring is harder because carry propagation couples all digit positions, and the system is globally constrained.

3 SMT Encoding

3.1 Bitvector Encoding in Z3

We encode the digit convolution constraints as assertions in the SMT-LIB bitvector theory, solved by Z3 [6].

Variables. Bitvector variables x and y of bit width $\beta/2$ (for balanced semiprimes).

Core constraint. Assert $x \cdot y = n$ using native bitvector multiplication, with extended width to avoid overflow.

Digit constraints. For a chosen base b, assert for each digit position k:

$$(x \cdot y) \bmod b^{k+1} = n \bmod b^{k+1}$$

using bitvector modular arithmetic.

3.2 Base as a Tunable Parameter

A distinctive feature of our encoding is that the base b is a free parameter with no analog in circuit-based SAT approaches. When $b = 2^k$, the operation $m \bmod b$ reduces to a bitvector extract (bit-slicing), which Z3 handles with zero overhead. Non-power-of-2 bases require integer division within the solver.

4 Experimental Results

4.1 Base Sweep

We tested 18 bases (2–512) on balanced semiprimes at 20–32 bits. Results at 32 bits:

Powers of 2 dominate due to Z3's bit-extraction optimization. Non-power-of-2 bases can *hurt* performance. The hypothesis optimal base $\sim 2^{\beta/4}$ is rejected ($R^2 = 0.09$).

4.2 Multi-Instance Statistics

50 balanced semiprimes per bit size (seeds 0–49), bit sizes 16–32. Three SMT configurations: raw, base-16, base-256.

At 32 bits, none of the digit-level encodings provide a statistically significant speedup (Cohen's $d < 0.31$). Classical algorithms (Pollard's rho, median 0.107 ms) outperform the best SMT configuration by approximately 12,000×.

Table 1: Base sweep results at 32 bits (seed=42).

Base	Runtime (s)	Speedup vs. raw
512	0.13	16.4×
16	0.34	6.5×
64	0.35	6.4×
128	1.33	1.7×
8	1.38	1.6×
raw	*2.21*	1.0×
5	5.72	0.39×

Table 2: Statistical significance of encoding differences at 32 bits.

Comparison	$\Delta\mu$ (s)	Cohen's d	Significant?
Raw vs. Base-16	0.13	0.12	No
Raw vs. Base-256	−0.19	0.19	No
Base-16 vs. Base-256	0.32	0.31	No

4.3 Scaling Laws

Extended sweep at 20–40 bits with power-of-2 bases. Fitting time $= a \cdot 2^{b \cdot \beta}$:

Table 3: Exponential scaling fits.

Configuration	Exponent b	R^2	Doubles every
Raw	0.339	0.688	2.9 bits
Base 64	0.378	0.959	2.6 bits
Base 32	0.458	0.971	2.2 bits

Digit constraints do not change the scaling class. They provide constant-factor speedups, more predictable runtime ($R^2 > 0.95$ vs. 0.69), and the ability to solve instances the raw solver cannot (40-bit).

4.4 Leaked Bits

Phase transition in solvability as a function of leaked bit fraction:

At 64 bits, our 50% threshold matches Ajani et al. [1]. At 128 bits, we require 70% vs. their ∼60%, suggesting circuit-based encodings produce more unit-propagation opportunities at larger sizes.

5 Lattice Structure and the Rank-1 Barrier

5.1 Carry Propagation as a Linear System

Introduce linearized variables z_{ij} intended to represent $x_i \cdot y_j$, and carry variables t_k. The constraints (5) become:

$$\sum_{i+j=k} z_{ij} + t_{k-1} - b \cdot t_k = c_k, \quad k = 0, \ldots, d-1 \tag{6}$$

This is a system of d linear equations in $m = d_x d_y + d$ integer unknowns: $A\mathbf{v} = \mathbf{c}$.

Table 4: Minimum leak fraction for successful factoring.

Semiprime bits	Factor bits	Min leak fraction
32	16	0.0
48	24	0.3
64	32	0.5
80	40	0.6
96	48	0.6
128	64	0.7

5.2 Lattice Formulation

The integer solutions form an affine lattice $\Lambda_n = \{\mathbf{v} \in \mathbb{Z}^m : A\mathbf{v} = \mathbf{c}\}$ of dimension $d_x d_y = \Theta(d^2)$.

5.3 The Rank-1 Constraint

For a lattice point to represent a valid factorization, the matrix $Z = (z_{ij})$ must satisfy:

$$\operatorname{rank}(Z) = 1, \quad \text{i.e.,} \quad Z = \mathbf{x} \otimes \mathbf{y} = \mathbf{x}\mathbf{y}^T \tag{7}$$

for $\mathbf{x} \in \{0, \ldots, b-1\}^{d_x}$, $\mathbf{y} \in \{0, \ldots, b-1\}^{d_y}$. This is a system of $\binom{d_x}{2}\binom{d_y}{2}$ quadratic equations (the 2×2 minors of Z), defining the Segre variety $\operatorname{Seg}(d_x, d_y)$ of dimension $d_x + d_y - 1$ in the ambient $d_x d_y$-dimensional space.

Within this representation, the rank-1 constraint is the precise locus of hardness. LLL efficiently handles the linear system (6). The rank-1 condition is the nonlinear obstruction.

5.4 Why LLL Works for Polynomial Factoring

Polynomial factoring over $\mathbb{Z}[x]$ has no carry propagation ($t_k = 0$). The constraint system is linear, and LLL finds factors as short lattice vectors. Integers lack the "dimension reduction" that polynomial degree provides: factoring a degree-d polynomial yields factors of strictly smaller degree, enabling recursion. No analogous property exists for integers.

5.5 SDP Relaxation Failure

We relaxed the rank-1 constraint via semidefinite programming, using ADMM-like alternating projection between the affine subspace $\{Z : A\mathbf{v} = \mathbf{c}\}$ and $\{Z : \operatorname{rank}(Z) = 1\}$.

Results. The integrality gap (fraction of spectral mass outside the leading singular value) is 30–39%, growing with problem size. The ADMM iteration oscillates without converging because the feasible set is non-convex. The alternating projection's apparent successes reduce to randomized trial division.

Assessment. Convex relaxation of the rank-1 constraint is insufficient for factoring.

5.6 Transfer Matrix and Carry Channel Analysis

The carry-propagation constraints (6) exhibit Markov chain structure. At each digit position k, define the *transfer matrix*

$$T_k[t_{\text{in}}, t_{\text{out}}] = \#\big\{(z_{ij})_{i+j=k} : \sum z_{ij} = c_k - t_{\text{in}} + b\, t_{\text{out}},\ 0 \le z_{ij} \le (b-1)^2\big\}$$

counting the bounded integer compositions. The total lattice point count is the matrix product evaluated at zero carries:

$$|\Lambda_n \cap \mathcal{B}| = \big(T_{d-1} \cdots T_1 \cdot T_0\big)[0, 0]$$

This replaces the heuristic estimate of Theorem 8(b) with a **rigorous, polynomial-time computation**: $O(d \cdot S^2)$ where $S = \max_k t_k^{\max}$.

Theorem 1 (Exact quadratic scaling constant)**.** *For base-b balanced semiprimes with $d_x = d_y = m$ and $d = 2m - 1$ digit positions:*

$$\log_2 |\Lambda_n \cap \mathcal{B}| = \frac{d^2}{4} + O(d)$$

for $b = 2$. For general base $b \geq 2$:

$$\log_2 |\Lambda_n \cap \mathcal{B}| = \frac{\log_2\big((b-1)^2 + 1\big)}{4} \cdot d^2 + O(d \log d)$$

In particular, $\alpha(2) = 1/4$, $\alpha(3) = \log_2 5/4 \approx 0.580$, $\alpha(10) = \log_2 82/4 \approx 1.588$.

The proof assembles two lemmas.

Lemma 2 (Row sum identity)**.** *Let T_k be the transfer matrix at position k with n_k product variables.*

(a) For base $b = 2$: every row sum of T_k equals $2^{n_k - 1}$ exactly.

(b) For general base b: every row sum of T_k equals $\big((b-1)^2 + 1\big)^{n_k}/b$ asymptotically, with relative error $O(1/n_k)$.

Proof. (a) Fix a carry-in t_{in}. The row sum $R(t_{\text{in}}) = \sum_{t_{\text{out}}} T_k[t_{\text{in}}, t_{\text{out}}]$ counts the total number of tuples $(z_{ij})_{i+j=k}$ with $z_{ij} \in \{0, 1\}$ (since $b = 2$ forces $(b-1)^2 = 1$) satisfying $\sum z_{ij} + t_{\text{in}} \equiv c_k \pmod 2$. This is the number of binary strings of length n_k with prescribed parity. By the binomial parity identity $\sum_{j \equiv r \pmod 2} \binom{n_k}{j} = 2^{n_k - 1}$ for any $r \in \{0, 1\}$, the row sum is $2^{n_k - 1}$ independent of t_{in} and c_k.

(b) For general b, each $z_{ij} \in \{0, 1, \ldots, (b-1)^2\}$. The row sum counts (z_{ij}) with $\sum z_{ij} \equiv c_k - t_{\text{in}} \pmod b$. By the DFT/roots-of-unity filter,

$$R(t_{\text{in}}) = \frac{1}{b} \sum_{\ell=0}^{b-1} \omega^{-\ell(c_k - t_{\text{in}})} \Big(\sum_{v=0}^{(b-1)^2} \omega^{\ell v} \Big)^{n_k}$$

where $\omega = e^{2\pi i/b}$. The $\ell = 0$ term gives $((b-1)^2 + 1)^{n_k}/b$. For $\ell \neq 0$, $|\sum_v \omega^{\ell v}| \leq (b-1)^2 + 1 - c$ for some $c > 0$ depending on b, so the remaining terms are exponentially smaller. □

Lemma 3 (Balanced sum)**.** *For balanced semiprimes with $d_x = d_y = m$ and $d = 2m - 1$:*

$$\sum_{k=0}^{d-1} n_k = m^2 = \frac{d^2}{4} + O(d)$$

where $n_k = \min(k+1,\ m,\ d-k)$ is the number of product variables at position k.

Proof. The triangular profile n_k rises from 1 to m (for $k = 0, \ldots, m-1$) and falls from m to 1 (for $k = m, \ldots, 2m-2$). The sum is $2\sum_{j=1}^{m-1} j + m = 2 \cdot \frac{(m-1)m}{2} + m = m^2$. Since $d = 2m - 1$, we have $m = (d+1)/2$ and $m^2 = d^2/4 + d/2 + 1/4$. □

Proof of Theorem 1. By Lemma 2(a), the row sums of each T_k are exactly $2^{n_k - 1}$ (base 2). Since the Perron–Frobenius eigenvalue of a non-negative matrix with constant row sums equals that row sum, $\rho(T_k) = 2^{n_k - 1}$. The total lattice point count satisfies

$$\log_2 |\Lambda_n \cap \mathcal{B}| = \sum_{k=0}^{d-1} \log_2 \rho(T_k) + O(d) = \sum_{k=0}^{d-1} (n_k - 1) + O(d)$$

The correction $O(d)$ accounts for boundary effects (the initial and terminal carry states are fixed at zero rather than summed freely). By Lemma 3,

$$\sum_{k=0}^{d-1}(n_k - 1) = m^2 - d = \frac{d^2}{4} - \frac{d}{2} + O(1)$$

yielding $\log_2 |\Lambda_n \cap \mathcal{B}| = d^2/4 + O(d)$, i.e., $\alpha = 1/4$.

For general base b, Lemma 2(b) gives $\log_2 \rho(T_k) = n_k \log_2((b-1)^2 + 1) - \log_2 b + O(1/n_k)$. Summing:

$$\log_2 |\Lambda_n \cap \mathcal{B}| = \log_2\big((b-1)^2 + 1\big) \cdot m^2 - d \log_2 b + O(d) = \frac{\log_2((b-1)^2 + 1)}{4} \cdot d^2 + O(d \log d) \quad \square$$

Spectral structure. The per-position entropy rate $h_k = \log_2 \rho(T_k) = n_k - 1$ (base 2) exhibits a universal *diamond profile*: rising from $h_0 = 0$ to $h_{d/2} = m - 1 \approx d/2$ and falling symmetrically, with total $\sum h_k = d^2/4 + O(d)$ as proven above. For base 2, the Diaconis–Fulman theory [5] shows the eigenvalues of T_k (normalized) are $1, \frac{1}{2}, \frac{1}{4}, \ldots, \frac{1}{2^{n_k-1}}$, with spectral gap $\frac{1}{2}$.

5.7 Carry Channel Capacity Bound

We formalize the information content of the carry chain via the forward–backward algorithm (HMM inference on the Markov chain). Let Z be a uniform random lattice point in $\Lambda_n \cap \mathcal{B}$ and $T(Z) = (t_0, \ldots, t_{d-1})$ its carry sequence.

Theorem 4 (Carry channel capacity)**.** *For base-b balanced semiprimes with d digit positions:*

(a) The carry chain entropy (mutual information between lattice point and carries) satisfies $H(T) = \Theta(d \log d)$. More precisely,

$$H(T) = \sum_{k=0}^{d-1} H(t_k \mid t_{k-1}) \le \sum_{k=0}^{d-1} \tfrac{1}{2} \log_2(2\pi e\, n_k/12) + O(d)$$

where n_k is the number of z-variables at position k.

(b) The residual uncertainty is $H(Z \mid T) = \Theta(d^2)$:

$$H(Z \mid T) = \log_2 |\Lambda_n \cap \mathcal{B}| - H(T) = \Theta(d^2) - \Theta(d \log d) = \Theta(d^2)$$

(c) The fraction of information revealed by the carries is $O(\log d/d) \to 0$.

Proof sketch. Part (a): By the Diaconis–Fulman spectral analysis [5], the conditional entropy $H(t_k \mid t_{k-1})$ equals the entropy of the Eulerian number distribution $A(n_k, j)/n_k!$, which is $\frac{1}{2}\log_2 n_k + O(1)$ by the CLT for descents in random permutations. Summing over the triangular profile: $\sum_{k=0}^{d-1} \frac{1}{2} \log_2 n_k = \log_2((d/2)!) = \frac{d}{2}\log_2(d/2) - \frac{d}{2}\log_2 e + O(\log d)$.

Part (b): Direct subtraction using Theorem 1. $\square$

Interpretation. The carry chain is a "thin channel" connecting the linear and nonlinear worlds: it transmits $\Theta(d \log d)$ bits from a $\Theta(d^2)$-bit lattice. Even with *perfect* carry knowledge, $2^{\Theta(d^2)}$ lattice points remain, of which exactly 2 are rank-1.

5.8 SOS Hierarchy and Rounding Hardness

We applied the Lasserre/SOS hierarchy at degrees 2 and 4. Degree-4 SOS closes the continuous relaxation gap to $< 0.1\%$ in 95% of instances—the moment matrix "knows" the factorization. Yet factor recovery succeeds only 26% of the time.

Theorem 5 (Sequential rounding hardness)**.** *For a d-digit base-b semiprime, any sequential rounding scheme (processing digits in order, no backtracking) using the degree-4 SOS moment matrix succeeds with probability at most*

$$P(success) \leq \prod_{k=0}^{d-1} \frac{1}{|F_k|} \leq 2^{-\Theta(d^2)}$$

where $|F_k|$ is the number of carry-compatible z-value tuples at position k.

Proof sketch. At each position k, given the correct carry-in t_{k-1}, there are $|F_k| = C(n_k \cdot (b-1)^2/2,\, n_k,\, (b-1)^2)$ valid z-value tuples (bounded compositions of a target into n_k parts). Only one tuple matches the true factorization. The per-step success probability is $1/|F_k|$. Since $|F_k|$ peaks at $\binom{n_k}{\lfloor n_k/2 \rfloor} \approx 2^{n_k}$ for the middle positions, the product is $2^{-\sum_k \log_2 |F_k|} = 2^{-\Theta(d^2)}$. □

Certification vs. search. The SOS gap measures *certification*: how well the relaxation certifies that a solution exists. Rounding measures *search*: extracting the solution from the certificate. Our result demonstrates a $\Theta(d^2)$-bit gap between certification (nearly perfect) and sequential search (exponentially hard). To our knowledge, this is the first such demonstration for a number-theoretic recovery formulation, paralleling results for planted clique [2] and tensor PCA [11].

Remark 1. *Theorem 5 applies to* sequential *rounding procedures that fix digits in order without backtracking. It does not rule out all polynomial-time extraction procedures from SOS relaxations—for instance, rounding schemes that exploit global correlations in the moment matrix or process digits in a non-sequential order. Extending the lower bound to arbitrary polynomial-time rounding remains an open problem; see Section 8.3 for discussion.*

5.9 Base-Dependent Spectral Landscape

We examine the singular value structure of lattice points in $\Lambda_n \cap \mathcal{B}$, reshaped as $d_x \times d_y$ matrices. A lattice point is *near-rank-1* if its second singular value satisfies $\sigma_2/\sigma_1 < 0.1$.

Theorem 6 (Base-dependent spectral landscape)**.** *For balanced semiprimes with d digit positions:*

(a) For base $b = 2$, the only lattice points with $\sigma_2/\sigma_1 < 0.1$ are the two exact factorizations. Zero spurious near-rank-1 points exist.

(b) For base $b = 10$, up to 79% of lattice points in $\Lambda_n \cap \mathcal{B}$ are spurious near-rank-1 ($\sigma_2/\sigma_1 < 0.1$ but not exact factorizations).

(c) Nuclear norm minimization over the carry-propagation lattice therefore finds the correct factorization in base 2, but returns misleading solutions in base 10.

Proof sketch. (a) In base 2, each $z_{ij} \in \{0, 1\}$, and the box constraints $z_{ij} \in \{0, (b-1)^2\} = \{0, 1\}$ force every lattice point to be a binary matrix. A binary matrix with $\sigma_2/\sigma_1 < 0.1$ must have rank 1 (up to rounding of a single entry): the Eckart–Young theorem shows the best rank-1 approximation captures $> 99\%$ of Frobenius norm, leaving $< 1\%$ for σ_2. With integer entries in $\{0, 1\}$, this forces exact rank 1.

(b) In base 10, $z_{ij} \in \{0, 1, \ldots, 81\}$. The large dynamic range permits many matrices where the leading singular direction dominates but $\text{rank}(Z) > 1$. Exhaustive enumeration at small d confirms up to 79% of lattice points are spurious near-rank-1.

(c) Nuclear norm $\|Z\|_* = \sum \sigma_i$ is the tightest convex relaxation of rank. Minimizing $\|Z\|_*$ subject to the carry-propagation constraints selects lattice points with concentrated singular values. In base 2, these are necessarily rank-1 (by part (a)). In base 10, the minimizer is generically one of the 79% spurious points. □

Interpretation. Base 2 is *spectrally clean*: the binary constraint $z_{ij} \in \{0, 1\}$ eliminates all spurious near-rank-1 points, making nuclear norm minimization exact. Larger bases introduce a "spectral fog" of near-rank-1 points that obscure the true factorization. This provides a structural explanation for why base 2 is special in the SOS analysis (degree-2 SOS is already tight due to $x_i(1 - x_i) \geq 0$).

5.10 Blind Deconvolution and SVD Recovery

The digit convolution formulation $Z = \mathbf{x} \otimes \mathbf{y}$ subject to linear constraints $A(Z) = \mathbf{b}$ is an instance of *blind deconvolution*: recovering two unknown signals from their convolution under linear measurements. We test SVD-based recovery: compute the minimum-norm solution Z^* to the carry-propagation system, take its SVD $Z^* = U\Sigma V^T$, and extract candidate factors from the leading left and right singular vectors.

Results.

- Average |correlation| between leading singular vectors and true factor digits: 0.48 (base 2), 0.83 (base 10).
- Factor recovery rate: 0% across all instances and bases tested.

Interpretation. The SVD "sees" the factors—the leading singular direction is correlated with the true factorization—but cannot extract them. This demonstrates the certification-vs-search gap at the spectral level: Z^* contains information about the factors (high correlation) but rounding the continuous singular vectors to valid digit sequences fails universally. The higher correlation in base 10 is explained by the larger digit range providing more "signal" per entry, but this is offset by the spurious near-rank-1 landscape (Theorem 6(b)).

5.11 Moment Indistinguishability

We provide empirical support for bounded-view rounding hardness by examining the degree-4 SOS moment matrix entries.

Proposition 7 (Moment collision frequency)**.** *At linear budget ($k = d$ moment entries selected by largest absolute value),* 26–43% *of randomly chosen semiprime pairs share identical degree-*4 *moment entries. The agreement fraction increases with digit count: from* 87% *at* $d = 8$ *to* 93% *at* $d = 16$.

Interpretation. Any rounding scheme that inspects $O(d^c)$ moment entries (for constant c) cannot distinguish a constant fraction of semiprime pairs. Since the correct rounding must depend on the specific semiprime, moment indistinguishability implies that $O(d^c)$-entry rounding fails on at least 26% of instances. The increasing agreement fraction with d supports the conjecture that the failure rate grows with problem size, consistent with the $2^{-\Theta(d^2)}$ sequential rounding bound of Theorem 5.

6 Restricted-Model Lower Bounds

6.1 Formal Model

Definition 1 (Digit Convolution Algorithm). *A procedure that factors n using: (i) free digit decomposition in any base b; (ii) free construction and polynomial-time manipulation of the carry-propagation lattice Λ_n; (iii) costly rank-1 oracle queries $\mathcal{O}_{\mathrm{R1}}$.*

Definition 2 (Rank-1 Oracle). *Given $Z \in \Lambda_n \cap \mathcal{B}$, the oracle returns:*

$$\mathcal{O}_{\mathrm{R1}}(Z) = \begin{cases} 1 & \text{if } \exists\, \mathbf{x}, \mathbf{y} \text{ s.t. } Z = \mathbf{x} \otimes \mathbf{y} \text{ and } 0 \le x_i, y_j \le b-1 \\ 0 & \text{otherwise} \end{cases}$$

A positive response yields the factorization.

6.2 Main Result: Rank-1 Sparsity

Theorem 8 (Rank-1 sparsity). *For a β-bit balanced semiprime $n = p \cdot q$ in base b with $d = \lceil \log_b n \rceil$:*

(a) The number of rank-1 integer matrices in $\Lambda_n \cap \mathcal{B}$ is at most 2.

(b) The ratio of rank-1 to total lattice points satisfies:

$$\frac{|\Lambda_n \cap \mathcal{R}_1 \cap \mathcal{B}|}{|\Lambda_n \cap \mathcal{B}|} \le 2^{-\Omega(d^2)}$$

which is proven rigorously by Theorem 1.

Proof. Part (a): A rank-1 matrix $Z = \mathbf{x} \otimes \mathbf{y}$ in $\Lambda_n \cap \mathcal{B}$ yields $p = \sum x_i b^i$, $q = \sum y_j b^j$ with $pq = n$. Since n is semiprime, at most two factorizations exist.

Part (b): The lattice point count satisfies $|\Lambda_n \cap \mathcal{B}| = 2^{d^2/4+O(d)}$ by Theorem 1, giving the ratio $\le 2/2^{d^2/4+O(d)} = 2^{-\Omega(d^2)}$. □

Interpretation. Rank-1 points are a $2^{-\Omega(d^2)}$ fraction of the feasible lattice. Any algorithm testing lattice points without exploiting rank-1 structure requires exponentially many queries.

6.3 Oracle Lower Bounds

Theorem 9 (Black-box lower bound). *In the black-box model, any deterministic algorithm requires $Q(\beta) \ge |\Lambda_n \cap \mathcal{B}|/2 - 1$ queries.*

Proof. Adversary argument: answer "no" until ≤ 2 candidates remain. □

Theorem 10 (Generic oracle lower bound). *Against a generic rank-1 oracle, any algorithm requires $Q(\beta) \ge 2^{\Omega(d^2/\log d)}$ queries in expectation to succeed with probability $\ge 2/3$.*

Theorem 11 (Conditional lower bound). *If FACTORING $\notin$ BPP, then any digit convolution algorithm with polynomial-time inter-query computation requires $Q(\beta) \ge \beta^{\omega(1)}$ rank-1 queries.*

Proof. By contrapositive: polynomially many polynomial-time queries yield a polynomial-time factoring algorithm. □

6.4 Barriers

Proving unconditional superpolynomial lower bounds in the full model faces three barriers: the algebraic structure of the Segre variety (smooth, rational, with explicit equations); the banded structure of the carry-propagation lattice enabling decomposition strategies; and the Razborov–Rudich natural proofs barrier [10].

7 Related Work

Circuit-based SAT factoring. Encoding multiplication as boolean circuits and converting to CNF [16, 12] handles ∼20–30 bit semiprimes. Our encoding operates at the arithmetic level, preserving number-theoretic structure.

Hybrid SAT + lattice. Ajani et al. [1] combine circuit-SAT with Coppersmith's method, factoring 768-bit numbers with ∼50% leaked bits. Our lattice analysis explains why: with partial knowledge, the remaining system is fully linear.

Quantum. Shor's algorithm [14] factors in polynomial time on quantum computers.

LLL and polynomial factoring. The LLL algorithm [7] achieves polynomial-time polynomial factoring because the coefficient constraints are linear—no rank-1 barrier.

8 Discussion and Future Work

8.1 The Framework

The central contribution is a *framework* decomposing factoring hardness into: (1) a tractable linear part (carry-propagation lattice, solvable by LLL); and (2) an intractable nonlinear part (rank-1 constraint, resistant to convex relaxation, exponentially sparse in the lattice).

8.2 Honest Assessment

We note candidly: digit constraints do not provide statistically significant speedup at 32 bits; the optimal base has no clean formula; SDP relaxation fails; the conditional lower bounds are trivially true; and classical algorithms dominate by 4+ orders of magnitude. We view these negative results as strengths, clarifying the boundaries of the approach.

8.3 Future Directions

1. **Non-sequential rounding hardness:** Theorem 5 covers sequential rounding. Extending to arbitrary polynomial-time rounding schemes (via pseudo-calibration [2] adapted to the digit convolution setting) would strengthen the certification-vs-search gap to an unconditional SOS lower bound.

2. **Bounded-view rounding hardness:** Proposition 7 provides empirical evidence that $O(d^c)$-entry rounding fails on a constant fraction of instances. Formalizing this as a lower bound—showing that any rounding scheme inspecting $o(d^2)$ moment entries fails with probability $1 - o(1)$—would strengthen the SOS separation. The increasing moment agreement fraction (87% → 93%) suggests an asymptotic argument may be feasible.

3. **Blind deconvolution rounding techniques:** The high SVD correlation (0.83 in base 10) suggests that more sophisticated rounding of singular vectors—e.g., Wirtinger flow, truncated power iteration, or semidefinite rounding of the leading singular subspace—could bridge the certification-vs-search gap in specific base regimes. Understanding which bases admit efficient spectral rounding is an open question.

4. **Intersection theory:** The intersection class of $\mathrm{Seg}(d_x, d_y)$ with carry-propagation lattices in the Chow ring could yield dimension bounds with complexity-theoretic implications. Bézout's theorem gives $|\Lambda_n \cap \mathrm{Seg}\,| \leq \binom{d-2}{d/2-1}$, which is tight at $d = 4$.

5. **Geometric Complexity Theory:** The rank-1 constraint has a natural interpretation in terms of $GL(d_x) \times GL(d_y)$ orbit structure, though the connection to the GCT program [9] remains distant and unexplored.

6. **Communication complexity:** Viewing factoring as a two-party problem ($\mathbf{x}$ with Alice, $\mathbf{y}$ with Bob) connects to known lower bounds for related problems.

Completed milestones. The analytical derivation of $\alpha = 1/4$ (previously listed as a future direction) is now resolved by Theorem 1 via the row-sum identity of Lemma 2.

Reproducibility

All code, data, and experiment scripts are available at `https://github.com/struktured-labs/factoring-lab`. The repository includes Python and Rust implementations, experiment scripts reproducing all results, and raw data in CSV format. Semiprimes are generated deterministically from integer seeds.

A Rust Port Performance

The backtracking solver was ported to Rust via PyO3/maturin, yielding 30–33× speedup with identical search paths (identical iteration counts):

Bits	Python (s)	Rust (s)	Speedup
12	0.00116	0.00004	26×
16	0.01033	0.00033	31×
20	0.14805	0.00452	33×
24	0.47351	0.01409	33×

The Rust implementation uses `u128` arithmetic, supporting semiprimes up to approximately 38 decimal digits.

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